

L14 - Quick Sort II

Thursday, April 30, 2020 12:29 PM

quickSort(Arr, l, r)

1. if $l < r$ {
2. let $p = \text{partition}(\text{Arr}, l, r)$
3. quickSort(Arr, l, $p-1$)
4. quickSort(Arr, $p+1$, r)

PARTITION(Arr, l, r)

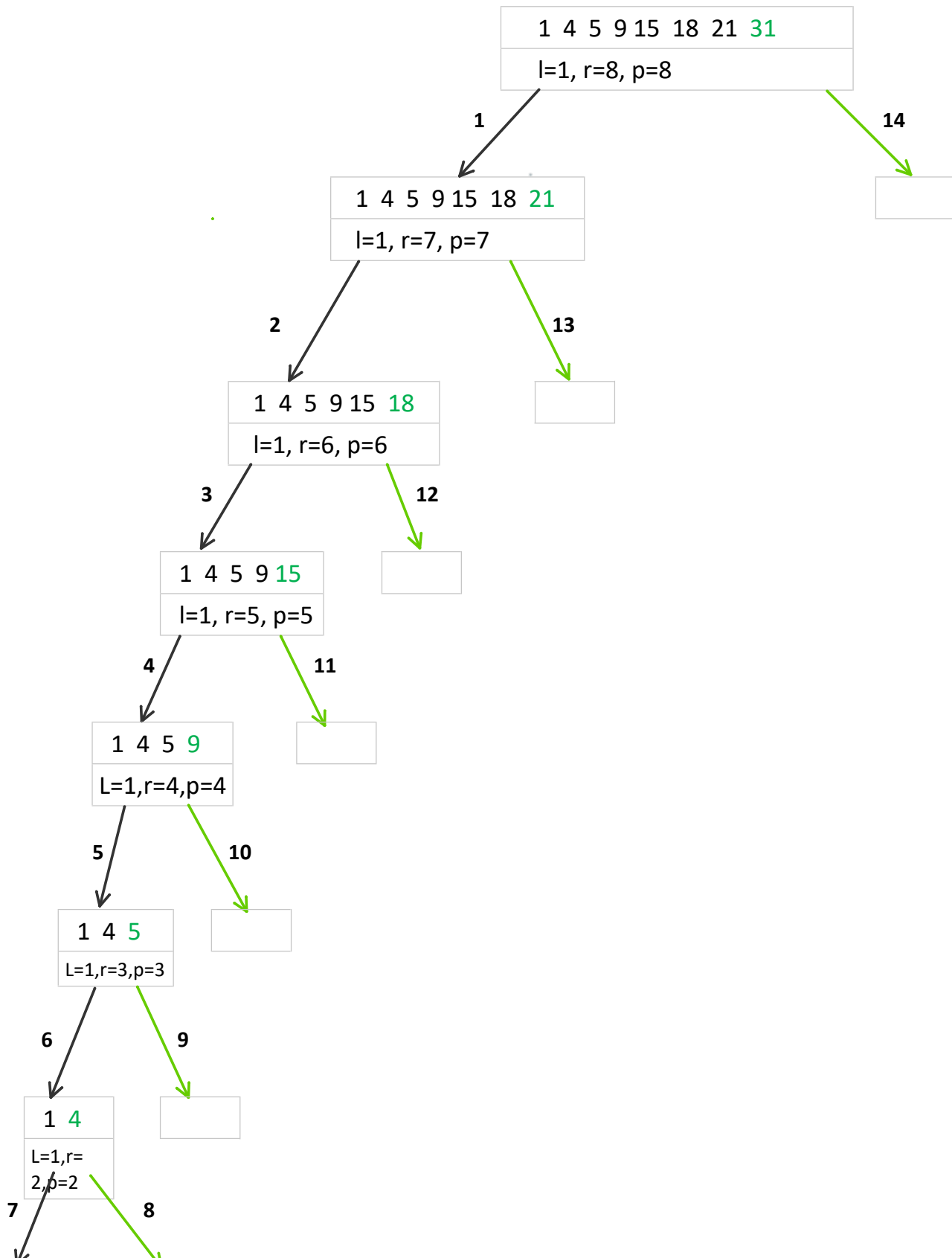
	Cost	Freq.
1. let pivot = Arr[r]	c1	1
2. let i = l - 1	c2	1
3. for j = l to r - 1 {	c3	$r-l+1$ (n)
4. if Arr[j] <= pivot {	c4	$r-l$ (n-1)
5. i = i + 1	c5	$r-l$ (n-1) (0)
6. exchange Arr[i], Arr[j]	c6	$r-l$ (n-1) (0)
7. }		
8. }		
9. exchange(Arr[i+1], Arr[r])	c9	1
10. return i + 1	c10	1

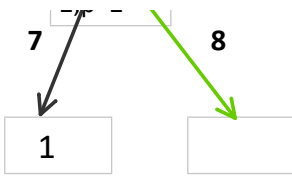
let $r - l + 1 = n$

$$\begin{aligned} T(n) &= (c3) * n + (c4+c5+c6) * (n-1) + (c1+c2+c9+c10) = (c3) * n + (c4+c5+c6) * n + (c1+c2+c5+c6) - (c4+c5+c6) \\ &= c10 * n + c11 \\ &= O(n) = \Omega(n) = \theta(n) \end{aligned}$$

Example:

Recursion Tree





Worst-Case Analysis

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5. }

Cost

Freq.

c_1

1

$O(n)$

1

(?1)

1

(?2)

1

$$\begin{aligned}
 T(n) &= (?1) + (?2) + O(n) + c_1 = (?1) + (?2) + \mathbf{cn} + c_1 = (?1) + (?2) + \mathbf{O(n)} \\
 &= T(n-1) + T(0) + O(n) = T(n-1) + c_1 + \mathbf{cn} \\
 &= T(n-1) + O(n)
 \end{aligned}$$

$$T(n) = \begin{cases} c_1; & \text{if } n \leq 1 \\ T(n-1) + O(n); & \text{if } n > 1 \end{cases}$$

$$T(n) = \begin{cases} c_1; & \text{if } n \leq 1 \\ T(n-1) + cn; & \text{if } n > 1 \end{cases}$$

where c & c_1 are positive constants:

Solving for

$$T(n) = T(n-1) + cn; \quad \text{if } n > 1 \quad (1)$$

Using equation 1, above, repeatedly,

$$T(n) = \underline{T(n-1)} + cn \quad (2)$$

Substituting $n-1$ for n in eq. 1 to get $T(n-1)$

Put the value of $T(n-1)$ in eq.2

$$= [T(n-2) + c(n-1)] + cn$$

$$= \underline{T(n-2)} + c(n-1) + cn \quad (3)$$

Substituting $n-2$ for n in eq. 1 to get $T(n-2)$

Put the value of $T(n-2)$ in eq.3

$$= [T(n-3) + c(n-2)] + c(n-1) + cn$$

$$= \underline{T(n-3)} + c(n-2) + c(n-1) + cn$$

...

$$= T(n-k) + c(n-(k-1)) + c(n-(k-2)) + \dots + c(n-2) + c(n-1) + cn \quad (4)$$

let $n-k=1$, therefore $k=n-1$,
substituting $k=n-1$ in equation 2

$$T(n) = T(n-n+1) + c(n-(n-2)) + c(n-(n-3)) + \dots + c(n-2) + c(n-1) + cn$$

$$= T(1) + c2 + c3 + \dots + c(n-2) + c(n-1) + cn$$

$$= c1 + c[2 + 3 + \dots + (n-2) + (n-1) + n]$$

$$= c1 + c [(n+2)(n - 1)/2]$$

$$T(n) = c1 + (c/2) [n^2 + n - 2]$$

$$= (c/2)n^2 + (c/2)n - c + c1$$

$$= \mathbf{O(n^2)}$$

Best-Case Analysis

$$T(n) = \begin{cases} c1; & \text{if } n \leq 1 \\ 2T(n/2) + O(n); & \text{if } n > 1 \end{cases}$$